Income Classification and Customer Segmentation Report

1. Executive Summary

This project uses Census-style demographic, employment, household, and financial-profile data to predict whether an individual earns more than $50,000. The dataset is highly imbalanced: only about 6.3% of records belong to the high-income class. Because of this imbalance, accuracy alone is not a reliable model evaluation metric.

After exploratory data analysis, feature engineering, and model training, the final XGBoost model achieved strong ranking performance:

Metric	Result
Test ROC-AUC	0.948
Test PR-AUC	0.655
Test Precision for > $50K	0.528
Test Recall for > $50K	0.670
Test F1 for > $50K	0.591
Selected Threshold	0.81

The final model was trained with both Census sampling weights and class-imbalance weights. The probability threshold was tuned on the validation set using positive-class F1 rather than relying on the default 0.5 threshold.

In addition to classification, customer segmentation was performed using engineered customer-profile features. Four clustering methods were compared: K-means, Gaussian Mixture Model, BIRCH, and Agglomerative Clustering. BIRCH achieved the highest silhouette score on the sampled data, while K-means was selected as the final segmentation model due to its scalability and interpretability.

2. Dataset Overview

The original dataset contained 199,523 rows and 42 columns. After removing exact duplicate rows, the final cleaned dataset contained 196,294 rows.

The target variable was converted into a binary label:

- 0: income <= $50,000
- 1: income > $50,000

Income Class	Count	Percentage
<= $50,000	183,912	93.69%
> $50,000	12,382	6.31%

This shows a strong class imbalance, so model evaluation focused on PR-AUC, precision, recall, and positive-class F1 rather than accuracy alone.

3. Data Cleaning

- Duplicate removal: Exact duplicate rows were removed from the dataset.
- Missing value handling: The only column with missing values was hispanic_origin. Missing values were filled with "?" to match the dataset's existing unknown-value convention.

- Target encoding: The original income label was converted into a binary target variable.
- Sampling weight separation: The Census weight column was removed from the feature set and stored separately as a sample weight. This avoids treating survey sampling weight as a predictive customer attribute.

4. Exploratory Data Analysis Findings

4.1 Class Imbalance

The high-income class is rare, representing only about 6.3% of the data. A model could achieve high accuracy by predicting most records as low income while still performing poorly on the business objective: identifying high-income individuals.

4.2 Age

Age shows a nonlinear relationship with income. Younger individuals and children are much less likely to be in the high-income group, while working-age adults are more represented among high-income records. Therefore, age was converted into age buckets instead of relying only on the raw numeric age.

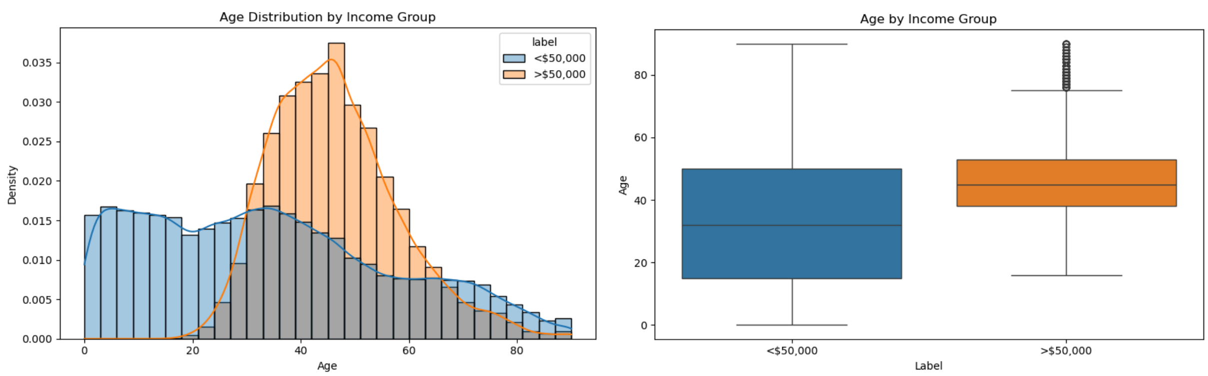

Fig.1 Age distribution (left) and boxplot (right) by income group.

4.3 Work Intensity

weeks_worked_in_year is one of the most important features. The low-income group contains many records with 0 weeks worked, while the high-income group is heavily concentrated around 52 weeks worked. This motivated the creation of work-intensity features such as worked_none_year, worked_full_year, and bucketed weeks-worked indicators.

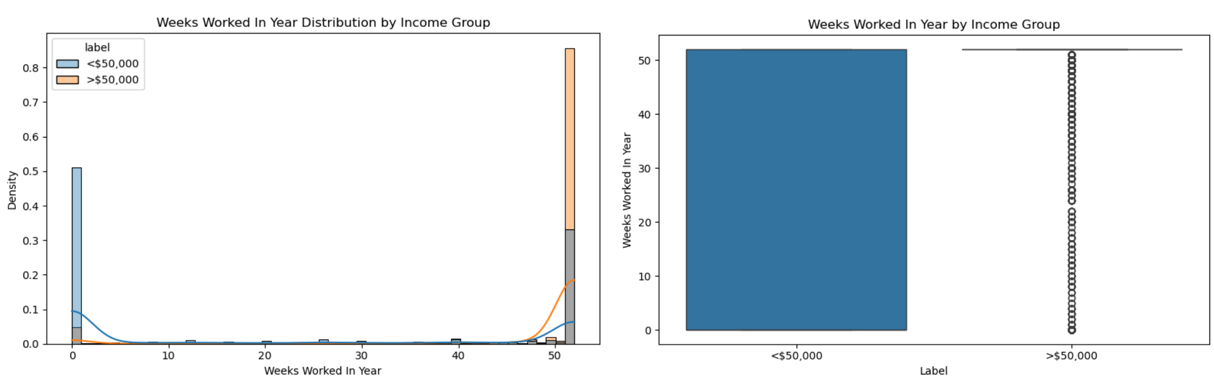

Fig.2 Weeks worked in year distribution (left) and boxplot (right) by income group.

4.4 Money and Investment Features

Money-related columns such as wage_per_hour, capital_gains, capital_losses, and dividends_from_stocks are highly zero-inflated and right-skewed. Both binary indicators and log-transformed versions were created so the model can capture whether a person has an income/investment source and the magnitude of that value.

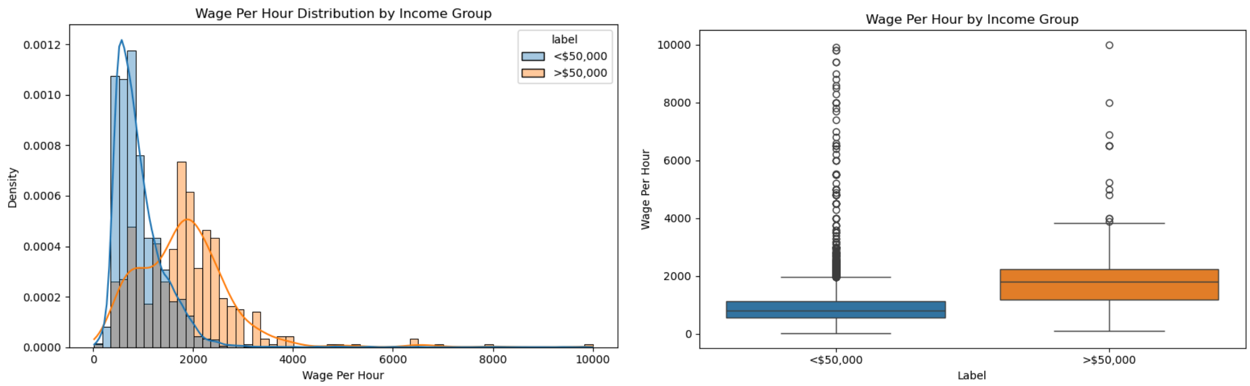

Fig.3 Money and Investment example: wage per hour distribution (left) and boxplot (right) by income group.

4.5 Categorical Features

Several categorical features showed strong separation between income groups, including education, marital_stat, class_of_worker, major_industry_code, major_occupation_code, full_or_part_time_employment_stat, tax_filer_stat, and detailed_household_summary_in_household. Other features were excluded because they were dominated by “Not in universe” / “?”, overly sparse, or redundant with stronger features. More details can be found in Section 5-7.

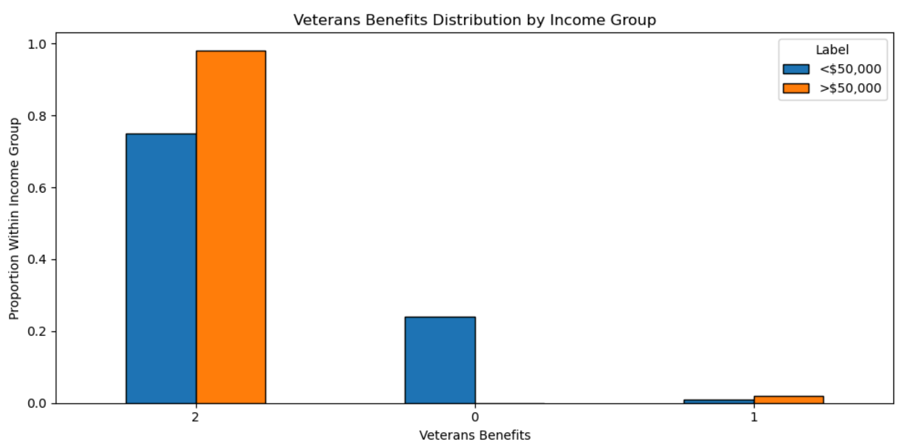

Fig.4 Categorical feature example: veterans Benefits classes by income group.

5. Sensitive Features

To reduce fairness and compliance risk, the following sensitive or sensitive-adjacent variables were excluded from final model training:

- race
- sex
- hispanic_origin
- country_of_birth_self

These variables may contain predictive signals, but using them directly in an income prediction model could introduce fairness concerns, especially in a financial-services context. They should be retained only for exploratory analysis or post-model fairness checks.

6. Feature Engineering

The final feature set contained 122 engineered features.

6.1 Age Features

The raw age variable was converted into one-hot encoded age buckets: 0-17, 18-24, 25-34, 35-44, 45-54, 55-64, and 65+. This captures nonlinear income patterns across life stages.

6.2 Money and Investment Features

For each major money-related feature, two engineered features were created: a binary indicator showing whether the value is positive and a log-transformed feature to reduce the impact of extreme values. Examples include has_dividends_from_stocks and log_dividends_from_stocks.

6.3 Weeks Worked Features

From weeks_worked_in_year, features were created for worked_none_year, worked_full_year, and bucketed work ranges. The original weeks_worked_in_year was also standardized and retained because it was highly predictive.

6.4 Industry and Occupation

The detailed industry and occupation recode features were excluded to reduce redundancy and sparsity. Instead, the broader major industry and occupation categories were one-hot encoded. Rare categories such as Armed Forces were removed or grouped due to extremely low frequency.

6.5 Grouped Categorical Features

Several categorical variables were grouped into broader categories before one-hot encoding. This reduced sparsity and improved interpretability. Examples include class_of_worker_grouped, marital_stat_grouped, education_grouped, full_or_part_time_employment_stat_grouped, family_members_under_18_grouped, and detailed_household_summary_in_household_grouped.

7. Features Excluded from Final Model

Feature	Reason
detailed_industry_recode	redundant with major_industry_code
detailed_occupation_recode	redundant with major_occupation_code
enroll_in_edu_inst_last_wk	mostly “Not in universe”; overlaps with age and education
reason_for_unemployment	overwhelmingly “Not in universe”
region_of_previous_residence	sparse and redundant with migration features
state_of_previous_residence	sparse and dominated by “Not in universe”
detailed_household_and_family_stat	overly granular; overlaps with household summary features
migration_code_change_in_msa	redundant with broader migration features
migration_code_move_within_reg	redundant migration signal
migration_prev_res_in_sunbelt	mostly unknown or not applicable
country_of_birth_father	high-cardinality and sensitive-adjacent
country_of_birth_mother	high-cardinality and sensitive-adjacent
fill_inc_questionnaire_for_veteran's_admin	overwhelmingly “Not in universe”

8. Model Training Approach

The data was split into train, validation, and test sets. The Census sampling weight was used as sample_weight rather than as a model feature. To address class imbalance, the training weight combined Census sampling weight and class-balanced weight.

Split	Rows	Percentage
Train	125,628	64%
Validation	31,407	16%
Test	39,259	20%

9. XGBoost Model

9.1 Baseline XGBoost

A baseline weighted XGBoost model was trained with n_estimators = 800, learning_rate = 0.03, max_depth = 4, and eval_metric = aucpr.

Metric	Value
Validation ROC-AUC	0.949
Validation PR-AUC	0.650

9.2 Hyperparameter Tuning

Optuna was used to tune XGBoost hyperparameters using validation PR-AUC as the objective. The best validation PR-AUC was 0.653.

Parameter	Best Value
learning_rate	0.0308
max_depth	5
min_child_weight	10
subsample	0.8876
colsample_bytree	0.8472
reg_lambda	1.6330
reg_alpha	2.9770

10. Threshold Tuning

Because the high-income class is rare, the default 0.5 threshold is not necessarily optimal. The validation set was used to choose the probability threshold that maximized positive-class F1.

Metric	Value
Threshold	0.81
Validation Precision	0.544
Validation Recall	0.676
Validation F1	0.603

Note: a **top 10% targeting** analysis was also performed. When targeting the top 10% highest-probability records, the model reached a probability cutoff of 0.742, **Precision@Top10%** of 0.465, and **Recall@Top10%** of 0.741.

11. Final Test Performance

The final tuned XGBoost model was evaluated once on the held-out test set using the selected validation threshold.

Metric	Test Result
ROC-AUC	0.948
PR-AUC	0.655
Precision for > $50K	0.528
Recall for > $50K	0.670
F1 for > $50K	0.591

Class	Precision	Recall	F1
<= $50K	0.98	0.96	0.97
> $50K	0.53	0.67	0.59

The model performs very well in ranking individuals by high-income likelihood, as shown by the high ROC-AUC and strong PR-AUC for an imbalanced dataset. The tuned threshold provides a reasonable tradeoff between precision and recall for the high-income class.

12. Model Interpretation with SHAP

SHAP was used to interpret the final XGBoost model. The most important individual features included:

- scaled_weeks_worked_in_year
- age_bucket_0_17
- tax_filer_stat_nonfiler
- detailed_household_summary_in_household_grouped
- scaled_num_persons_worked_for_employer
- scaled_dividends_from_stocks
- education_grouped_bachelors
- education_grouped_less_than_high_school
- education_grouped_graduate_degree
- scaled_capital_gains
- worked_full_year
- marital_stat_grouped_married
- major_occupation_code_adm_support_including_clerical

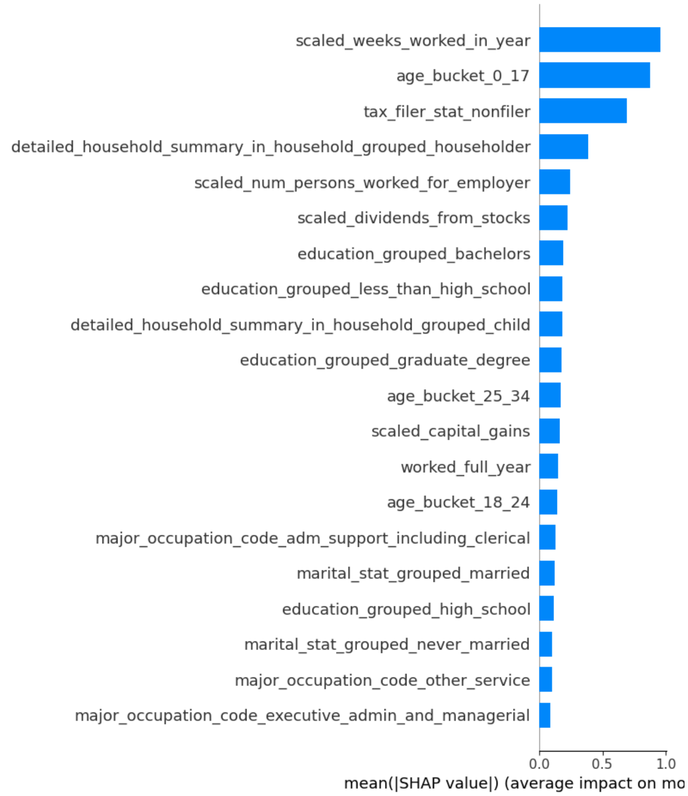

Fig.5 Feature importance ranking indicated by SHAP.

Feature Group	Interpretation
Age bucket	Life stage is highly predictive
Weeks worked	Work intensity is one of the strongest signals
Tax filer status	Filing status strongly separates income groups
Education	Higher education levels are associated with higher income
Major occupation	Occupation type strongly affects income likelihood
Household summary	Household role/status contributes predictive signal
Major industry	Industry group adds employment context
Marital status	Marital/family structure is associated with income differences
Dividends / capital gains	Investment income is informative for high-income prediction

Overall, the model relies most heavily on work intensity, age/life stage, tax filing status, education, occupation, household structure, and investment-related features.

13. Customer Segmentation

Customer segmentation was performed using the engineered feature matrix. The income label was not used to create the clusters, but it was used afterward to profile the segments. A 20,000-row sample was used for visualization and method comparison. The high-dimensional feature matrix was projected into lower-dimensional space using TruncatedSVD.

Method	Silhouette Score	Number of Clusters
BIRCH	0.430	4
K-means	0.315	4
Agglomerative Clustering	0.298	4

Gaussian Mixture Model	0.269	4

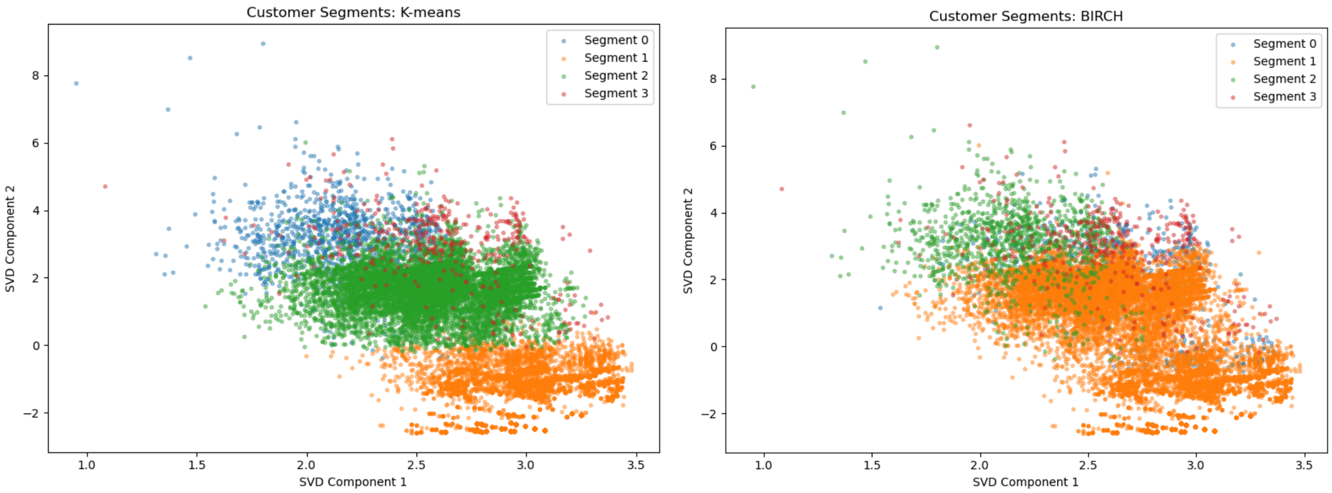

Fig.5 K-means (left) and BIRCH (right) customer segmentation visualized in reduced feature space

Although BIRCH had the highest silhouette score, K-means was selected as the final segmentation model because it is scalable, stable, and easier to explain in a business setting.

14. Segment Profiles

The final K-means model produced 4 customer segments. Segment names are business-friendly interpretations based on the segment profile statistics, not labels generated directly by the algorithm.

Segment 0: Older Non-Working / Low-Income Segment

Profile Attribute	Value
Average age	54.9
High-income rate	1.4%
Average weeks worked	1.4
Worked no weeks	88.8%
Top education	High school graduate
Top class of worker	Not in universe
Top marital status	Married-civilian spouse present

Interpretation: This segment appears to represent older individuals largely outside the active labor force. The high-income rate is very low, and work intensity is minimal.

Segment 1: Active Working Mainstream Segment

Profile Attribute	Value
Average age	38.8
High-income rate	11.3%
Average weeks worked	46.8
Worked full year	72.2%
Top class of worker	Private
Top industry	Retail trade
Top occupation	Administrative support / clerical
Top education	High school graduate

Interpretation: This segment represents active workers, mostly in private-sector roles. The high-income rate is above the overall dataset average but still moderate.

Segment 2: Children / Young Non-Working Segment

Profile Attribute	Value
Average age	8.1
High-income rate	0.0%
Average weeks worked	0.3
Worked no weeks	97.6%
Top education	Children
Top class of worker	Not in universe
Top marital status	Never married

Interpretation: This segment clearly represents children or very young individuals with no labor-force participation. It has essentially no high-income potential under the current income definition.

Segment 3: Investment / Capital-Loss High-Income Segment

Profile Attribute	Value
Average age	44.0
High-income rate	30.1%
Average weeks worked	41.6
Worked full year	67.2%
Has capital losses	100%
Has dividends from stocks	25.2%
Segment-level high-income rate	Highest among the four segments

Interpretation: This segment appears to capture financially active individuals with investment-related activity, especially capital losses and dividends. It has the highest high-income rate and may represent a strong target group for financial products.

15. Business Recommendations

Use Model Scores for Targeted Outreach

The model should be used as a ranking tool to prioritize individuals with high predicted probability of earning more than $50,000. For limited campaign budgets, the business can target the top 10% highest-scored records.

Prioritize High-Value Segments

Segment 3 has the highest high-income rate and strong investment-related characteristics. This group may be appropriate for wealth management, investment advisory, premium financial services, brokerage, or retirement planning campaigns.

Avoid Low-Value Targeting Groups

Segments 0 and 2 have very low high-income rates. These groups should receive lower priority for high-income-focused campaigns, though they may still be relevant for different products or life-stage strategies.

Use Sensitive Features Only for Fairness Checks

Sensitive variables should not be used for model prediction. However, they can be used after modeling to evaluate whether model performance varies meaningfully across demographic groups.

16. Limitations

- Income threshold is binary: The model predicts whether income is above or below $50,000, not actual income amount.
- Dataset is Census-style, not transaction-level: The dataset does not include balances, transaction history, product ownership, or behavioral engagement data.
- Class imbalance remains challenging: The high-income class is rare. Even with strong ROC-AUC, precision for the high-income class remains moderate.
- Some features may act as proxies: Even after excluding direct sensitive variables, other features may partially proxy for demographic differences. A fairness audit is recommended before production use.
- Segmentation is profile-based: The segments should be validated with downstream campaign outcomes before operational use.

17. Conclusion

The final XGBoost model provides strong predictive ranking performance for identifying high-income individuals in an imbalanced Census-style dataset. The model achieved a test ROC-AUC of 0.948 and PR-AUC of 0.655, with a tuned threshold that balanced precision and recall for the rare high-income class.

SHAP analysis showed that the strongest predictors were work intensity, age, tax filer status, education, occupation, household structure, and investment-related features. Customer segmentation further identified meaningful customer groups, including an active working segment and a high-value investment-related segment with the highest high-income rate.

Overall, the project demonstrates a complete workflow from EDA and feature engineering to imbalance-aware modeling, threshold tuning, model interpretation, and customer segmentation.